



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	ATMOSPHERIC PHYSICS				
Type of Course	Major Elective				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Basic thermodynamics. 2. Basic electrostatics.				
Course Summary	This course explores the structure and dynamics of the Earth's atmosphere. The vertical structure of the atmosphere, atmospheric thermodynamics, Earth's heat and radiation budget as well as atmospheric electricity are discussed. Basics of climate change and atmospheric photochemistry are also introduced.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the basic structure of the atmosphere and its constituents stratified to several layers. Understand rainfall, its distribution as well as the role played by winds.	U, R	F	Instructor-created exams / Quiz
CO2	Obtain basic idea of global warming. Apply the concepts of pressure, temperature, humidity to atmosphere and their role in climate change.	U, Ap	F, P	Instructor-created exams / Quiz
CO3	Apply thermodynamical concepts and latent energy to analyse stability of air parcel	Ap, An	F, C	Instructor-created exams / Quiz
CO4	Understand the atmospheric energy budget and the role played by radiation in it.	U, Ap	F	Instructor-created exams / Quiz
CO5	Understand basic atmospheric photochemistry and the role of trace gases.	U	F	Instructor-created exams / Quiz
CO6	Understand cloud physics and thunderstorm electricity. Apply the concept of electric field to atmosphere in the form of lightning and learn about lightning protection measures.	U, Ap	F, P	Instructor-created exams / Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	INTRODUCTION TO ATMOSPHERIC PHYSICS		13	18
	1	The atmosphere — Origin and Composition of the atmosphere.	2	
	2	Different layers of the atmosphere. Vertical thermal structure of the atmosphere – distribution of pressure and temperature.	3	
	3	Global distribution of precipitation.	2	
	4	Measurement techniques: air temperature, relative humidity, pressure, rain fall.	2	
	5	Introduction to atmospheric boundary layer.	2	
	6	Greenhouse effect, global warming.	2	
<i>Atmospheric Physics</i> , Chapter 1, Chapter 2: 2.1-2.4, Chapter 3: 7. <i>Basics of Atmospheric Science</i> , Chapter 1:1.3, 1.5.1, Chapter 8: 8.1, 8.2, 8.9. 8.10				
II	ATMOSPHERIC THERMODYNAMICS		9	12
	7	Adiabatic processes –concept of an air parcel, lapse rate, thermodynamic parameters and diagrams.	3	
	8	Atmospheric stability- unsaturated air, saturated air, conditional and convective instability.	3	
	9	CAPE, CINE, CIFK and CISK.	3	
Chapter 3: 3.4, 3.6. of <i>Atmospheric Science</i> , John M. Wallace. Peter V. Hobbs, Chapter 5: 5.5.3. of <i>Atmospheric Science</i> ,				
III	ATMOSPHERIC RADIATION AND PHOTOCHEMISTRY		15	25
	10	Radiation: The spectrum of radiation – Black body radiation.	1	
	11	Planck function, radiative properties of non-black bodies.	3	

	12	Scattering and absorption by air molecules and particles.	2	
	13	Atmospheric windows, solar constant.	2	
	14	Surface radiation budget and net radiation, radiative forcing.	3	
	15	Atmospheric photochemistry of NO, NO ₂ , O ₃ , CH ₄ , CO.	3	
	16	Absorption of radiation by trace gases.	1	
<i>Atmospheric Physics</i> , Chapter 3: 2, 3, 8, 10. <i>Basics of Atmospheric Science</i> , Chapter 4: 4.1, 4.2, 4.3, 4.4, 4.5, Chapter 12: 12.1.2, 12.13, 12.1.4. <i>Atmospheric Chemistry and Physics</i> , Chapter 4: 4.5, 4.6, 4.7, 4.9, 4.10.				
IV	ATMOSPHERIC ELECTRICITY		11	15
	17	Cloud morphology, structure and dynamics of thunder clouds.	2	
	18	Fair weather electric field in the atmosphere and potential gradient.	2	
	19	Ionisation in the atmosphere, conduction currents, point discharge current, air Earth currents.	2	
	20	Electric field in thunderstorm, theories of thundercloud electrification.	2	
	21	Lightning discharge, global electric circuit, Cloud electrification mechanisms.	2	
	22	Physics of lightning-lightning protection.	1	
<i>Atmospheric Physics</i> , Chapter 6: 2-7. <i>Atmospheric Science</i> , Chapter 5: 5.5.2, 5.5.3. Chapter 6: 6.7, Chapter 8: 8.3.2.				
V	OPEN ENDED MODULE		12	
		Optical features of the atmosphere: Refraction, scattering, Diffraction phenomena, aurorae, Indian monsoon.		
	Relevant sections from chapters 2-7, <i>Rainbows, Halos and Glories</i> .			

Books and References:

1. Atmospheric Physics, J. V. Iribarne, H.R. Cho, Springer, 1980 (Book 1)
2. Atmospheric Science, John M. Wallace. Peter V. Hobbs, Elsevier, 2006 (Book 2)
3. Atmospheric Chemistry and Physics, John H. Seinfeld, Spyros N. Pandis, John Wiley & Sons, 2006.
4. Basics of Atmospheric Science 2nd Edition, A. Chandrasekar, PHI, 2010
5. Rainbows, Halos and Glories, Robert Greenler, Cambridge University Press, 1980.
6. Atmosphere, Weather and Climate 9th edition, Roger G. Barry, Richard J Chorley, Routledge, 2017.

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO3	PS O4	PSO 5	PSO 6	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO7
CO 1	3	2	2	2	3	3	3	2	3	1	2	3	3
CO 2	2	2	2	2	3	2	3	2	3	1	2	3	3
CO 3	2	3	2	3	3	3	3	2	3	1	2	3	3
CO 4	2	2	3	2	2	2	3	2	3	1	2	3	3
CO 5	2	2	2	2	3	2	3	2	3	1	2	3	3
CO 6	2	3	3	2	2	3	3	2	3	1	2	3	3

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	